Reg. No. | | | | | | | | | | | [14CE62]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE – 2019
RAILWAY AND AIRPORT ENGINEERING

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions selecting atleast **TWO** from each part.
2. Missing data may be suitably assumed.

PART - A

1. a. Define gauge of a railway track. Enumerate different gauges used in India. **04**

   b. What do you understand by permanent way? Draw a typical cross section of a B.G. track in Embankment on straight track. **10**

2. a. Find out the expression for sleeper density for a B. G. track if 19 sleepers are used under a rail length. **03**

   b. Calculate the steepest gradient on a straight track using the following data, for a train having 20 wagons. **07**

   i. weight of each wagon = 20 tonnes,

   ii. Rolling resistance of wagon = 2.5kg/tonne,

   iii. Speed of train = 50kmph,

   iv. Tractive effort of locomotive = 12 tonnes,

   v. Rolling resistance of locomotive = 3.5 kg/tonne.

   vi. Weight of locomotive with tender = 120 tonnes

   c. List the functions of sleepers and ballast. **04**

3. a. What is creep? Discuss the theories propounded to explain probable causes of creep(any 2) **07**

   b. On a B.G. 3^0 curve, the "equilibrium cant" is provided for a speed of 70kmph. **04**

   i. Calculate the value of equilibrium cant

   ii. Allowing a maximum cant deficiency, what would be the maximum permissible speed on the track?

   c. Define super elevation and its objectives on curves of a railway track. **03**

4. a. State the various methods of plate laying and explain the methods of plate laying widely used in India. **10**

   b. Explain the necessity and advantages of good maintenance of track. **04**

PART - B

5. a. Explain the various factors considered for airport site selection. **06**

   b. Describe the procedure adopted to construct wind rose. **08**

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6. a. An airport is proposed at an elevation of 400m above MSL where the mean of maximum and mean of average daily temperature of the hottest month are 44.8 C and 26.2 C respectively. The maximum elevation difference along the proposed profile of runway is 6.3m. if the basic length of runway is 1260m, determine the actual length of the runway to be provided. **10**

b. Describe any two of the following terms; **04**

i. Cross wind component.

ii. Head wind.

iii. Wind coverage.

7. a. What are the functions of Hangars? Explain its types. **08**

b. Draw the neat sketch of airport layout of single runway and two parallel runway, indicating its component parts. **06**

8. a. Explain the following with the help of neat sketch;

i. Runway lighting **07**

ii. Taxiway lighting **07**